Aaradhya Dev Tamrakar

Embedded Systems · Edge AI · Linux & Systems Engineering

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Final-year Electronics & Information Engineering undergraduate specializing in embedded AI, low-level systems programming, and deterministic control. Technical lead for SPARK (ESP32-S3 wearable fall detection running quantized 18.5 KB INT8 CNNs under FreeRTOS), developer of Windows DFIR stealth-evasion triage engines, and kernel-level NTFS privacy tooling. Experienced in C/C++, Win32 APIs, RTOS, and protocol verification.

EDUCATION

Bachelor of Engineering (B.E.) in Electronics, Communication and Information Engineering | December 2022 – Expected January 2027

Kathmandu Engineering College, Institute of Engineering (IOE), Tribhuvan University • Cumulative Average: ~75% across completed semesters

Relevant Coursework: Machine Learning · Signals and Systems · Digital Electronics · Control Systems · Communication Systems · Engineering Mathematics · Probability and Statistics · Object-Oriented Programming (C++) · Computer Networks · Microprocessors

FELLOWSHIPS & APPLIED ML EXPERIENCE

Fuse AI Fellow — Fusemachines AI Fellowship 2026 | 2026 – Present (Active)

RESEARCH & TECHNICAL PROJECTS

SPARK — Safety-Critical Wearable Edge-AI Fall Detection System | 2025 – Present

ESP32-S3 · C/C++ · TFLite Micro · PyTorch · Python · BLE · FastAPI · SHAP

- Technical lead for ML modeling, quantization, and gateway architecture in a four-person team; designed a two-tier fall detection system eliminating cloud latency and privacy hazards.
- Trained 1D-CNN on the **SisFall benchmark (38 subjects, 38,420 windows)**; achieved **0.9185 AUC-ROC (95% CI: [0.9065, 0.9305])** and **87.81% sensitivity** under subject-stratified 5-fold cross-validation with zero data leakage.
- Executed post-training integer quantization (PTQ), compressing model flash memory footprint to **18.5 KB** for deterministic, **sub-100ms** on-device confirmation on the **ESP32-S3** microcontroller.
- Engineered BLE/serial local gateway, local **SHAP feature attribution**, FastAPI REST event store, and automated clinical incident PDF report generation; verified by **56 host CTest/Unity unit tests** and **48 Python integration tests**.

Headless Invariant Assurance Engine (INV-BMK-001) | 2026

Python · Z3 · SMT-LIB2 · State Machines · Pytest · Formal Methods

- Developed a formal verification benchmark encoding reactive state-machine contracts into **SMT-LIB2/Z3**, inductively proving candidate safety invariants and generating executable counterexamples.
- Evaluated **12 safety properties across 4 state machines**, discovering and sandbox-replaying **4/4 planted violations (100% recall, 0.0% false-discovery rate)** with **10.844 ms mean solver latency**.

Fleet-Orchestrator — Multi-Account Autonomous Agent Swarm & Consensus Engine | 2026 – Present

Python 3.12+ · FastAPI · SQLite WAL · asyncio · MCP · CDP · Copilot CLI

- Architected an autonomous multi-agent coordinator managing background worker swarms across **31 Copilot accounts** with atomic **SQLite WAL lease state machines** and crash recovery.

Cyber-Forensics — Windows DFIR Stealth-Evasion Triage Engine | 2026

PowerShell 7 · Win32/NT APIs · NTFS Streams · DFIR Parsers · Security

- Built a high-performance Windows DFIR triage engine detecting hidden **NTFS Alternate Data Streams (ADS)**, **CLSID and Unicode RLO masquerades**, and **PE header anomalies**.
- Automated Prefetch execution artifact harvesting and maintained tamper-evident chain-of-custody evidence ledgers for rapid security triage.

Win-Vault — Kernel-Level NTFS Folder Privacy Engine | 2026

PowerShell 7 · Windows Forms · NTFS ACLs · Win32 API · SHA-256

- Engineered a zero-dependency Windows folder locking utility enforcing kernel-level access denial via native **NTFS Access Control Lists (ACLs)** and **SHA-256** password hashing.

Gesture-Controlled Self-Balancing Robot (GCSBR) | 2025

Arduino · C/C++ · MPU-6050 · PID Control · MATLAB · Android · MediaPipe

- Implemented real-time **PID stabilization firmware** on an inverted pendulum robot, validated via a MATLAB-simulation-to-hardware parameter transfer pipeline.
- Developed an Android companion app (**Kotlin**, **MediaPipe Hands**, **CameraX**, **Bluetooth**) mapping hand gestures to directional actuation commands in real time.

Edge AI Stability Detection System | 2025

Python · scikit-learn · FastAPI · Joblib · IMU Data

- Serialized the pipeline via Joblib and deployed an ultra-low-latency **FastAPI REST inference endpoint** for robotics integration.

Win-Governor / System-Optimizer — Workstation Memory Reclamation Engine | 2025 – 2026

C# .NET 10 · WPF · Win32/NT APIs · Windows Internals

- Built a native Windows performance utility reclaiming **1.2–3.4 GB working-set RAM** under heavy developer workloads via undocumented Win32/NT APIs.

Utility MCP Suite — LocalSend, Google Classroom & Fusion 360 Bridges | 2026 – Present

Node.js ESM · Python · MCP Protocol · REST / HTTP · mTLS

- Developed **localsend-mcp** enabling AI agents to discover LAN peers and transfer multi-gigabyte build artifacts over secure mTLS (<25ms transport latency).

LEADERSHIP & SERVICE

Vice Chair (2026 – Present); Vice Secretary (2025 – 2026) — IEEE KEC KTM Student Branch | 2025 – Present

- Direct branch strategy, executive committee operations, technical workshops, and peer learning initiatives for engineering students.
- Previously managed operational documentation and workshop logistics across department-level technical events as Vice Secretary.

Event Manager (2026 – Present); Resource Manager (2024 – 2026) — Electronics Project Club (EPC), Kathmandu Engineering College | 2024 – Present

- Plan and execute technical hackathons, project exhibitions, and hardware workshops; manage lab equipment allocation across engineering departments.

Makerspace Ambassador — KEC Maker's Space, Kathmandu Engineering College | June 2026 – Present

- Support student rapid prototyping, 3D printing, and PCB bring-up; maintain lab website infrastructure and asset consistency.

TECHNICAL SKILLS

Programming Languages: Python, C, C++, SQL (PostgreSQL, SQLite), Kotlin, C# (.NET), VHDL, PowerShell, Bash, JavaScript / TypeScript

Edge AI & Embedded Systems: ESP32-S3 (ESP-IDF / FreeRTOS), TensorFlow Lite Micro (INT8 PTQ), Arduino, MPU-6050, UART / I2C / SPI, Bluetooth Low Energy (BLE), MATLAB / Simulink

Formal Methods & Verification: Z3 SMT Solver, SMT-LIB2, State-Machine Contract Verification, Inductive Invariant Proofs, Counterexample Generation & Sandbox Replay

AI Systems, Backend & MLOps: FastAPI, Docker, Model Context Protocol (MCP), asyncio, MLflow, Evidently AI, Airflow, PostgreSQL, SQLite FTS5, REST APIs

Systems Security & Forensics: Windows Win32/NT APIs, NTFS Alternate Data Streams (ADS), NTFS ACLs, PE Header Verification, Prefetch Forensics, DFIR Triage

Engineering Practice & Tools: Git, GitHub, Linux (Ubuntu), Jupyter, Google Colab, LaTeX / Overleaf, WordCraft, Pytest, CMake / CTest, Automated CI/CD